

Continuous functions – 4

Uniform continuity and the modulus of continuity

Homework

1 сентября 2026 года

Theoretical minimum. Uniform continuity and its sequential criterion; the modulus of continuity and its subadditivity; Cantor's theorem; Lipschitz and Hölder conditions; McShane's extension theorem; the linear growth bound; the behaviour of uniform continuity under sums, products and compositions.

1. Prove that ω_f is non-decreasing and subadditive, that f is uniformly continuous if and only if $\omega_f(0^+) = 0$, and that for a concave modulus the ratio $\frac{\omega_f(\delta)}{\delta}$ is non-increasing.
2. Prove that a continuous f on $[0, \infty)$ with a finite limit at $+\infty$ is uniformly continuous, and that a periodic continuous function on $\mathbb{R}$ is uniformly continuous.
3. Prove that a differentiable function with bounded derivative is uniformly continuous, and give an example of a uniformly continuous function whose derivative is unbounded.
4. Prove that a Hölder condition $|f(x) - f(y)| \leq C|x - y|^\alpha$ with $\alpha > 1$ forces f to be constant, while for every $\alpha \in (0, 1)$ there are non-constant examples.
5. (S.T. Yau CSMC 2018) Let $A \subseteq \mathbb{R}^m$ and $f: A \rightarrow \mathbb{R}$. Suppose there is $\alpha > 0$ such that for every point z of the graph $G = \{(x, f(x))\}$ the cone $C_{z,\alpha} = z + \{(y_1, y_2) : |y_2| > \alpha|y_1|\}$ misses G . Prove that f is Lipschitz with constant α .
6. Prove that the sum of two uniformly continuous functions is uniformly continuous, that the product need not be, and that the composition of uniformly continuous functions is uniformly continuous.

7. Prove that f is uniformly continuous on A if and only if it maps Cauchy sequences of A to Cauchy sequences; deduce Cantor's theorem once more.
8. Construct a bounded continuous function on $\mathbb{R}$ which is not uniformly continuous, and an unbounded uniformly continuous one.
9. Let f be uniformly continuous on $\mathbb{R}$ and let $f_n(x) = f(x + a_n)$ with $a_n \rightarrow 0$. Prove that $f_n \rightarrow f$ uniformly on $\mathbb{R}$, and show that continuity alone is not enough.
10. Let $f: [0, \infty) \rightarrow \mathbb{R}$ be uniformly continuous and suppose $\int_0^\infty f$ converges. Prove that $f(x) \rightarrow 0$ as $x \rightarrow \infty$, and give an example showing that uniform continuity cannot be dropped.